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Getting Started with Portenta Max Carrier and Portenta X8

Get started with your Portenta Max Carrier and Portenta X8

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Introduction

Using the Portenta X8 with the Portenta Max Carrier enables a large suite of peripherals and features. With the Portenta X8 it is possible to use Linux containers and Python® scripts, it is also possible to use the M4 core on the board to interface with the Portenta Max Carrier. In this tutorial we will focus on the peripherals and features that are available to us.

Goals

The goals of this project are:

Learn how to connect the Portenta X8 and Portenta Max Carrier

Learn about the Portenta Max Carrier's headers, peripherals and connectors

Required Hardware and Software

[Arduino® Portenta X8](#)

[Arduino® Portenta Max Carrier](#)

[USB-C® cable \(USB-C® to USB-A cable\)](#)

The Portenta Max Carrier

The Arduino® Portenta Max Carrier was designed to add capabilities to the Arduino® Portenta X8 board and provide easy access to its onboard peripherals. It is designed to enable research and development of industrial-grade advanced applications, from fast prototyping to deployable robust single-board computers.

To make use of the Portenta Max Carrier you will need to

a 3.7V 18650 Li-Ion battery, connected to the Portenta Max Carriers battery clips.

Function	Supported on the Portenta X8 M4 Core	Supported on Portenta X8's NXP® i.MX 8M Mini Processor
USB Host	USB 1.0	USB 2.0
Ethernet	Fast Ethernet	1 Gbps
CAN	Yes	Yes
Mini PCIe (USB)	USB 1.0	USB 2.0
Mini PCIe (PCIe)	No	PCIe 2.0
Battery Charger	Yes	Yes
LoRa® technology	Yes (additional software required)	Yes (additional software required)
NB IoT/CatM1/2G	Yes (additional software required)	Yes (additional software required)
Camera	No	MIPI up to 4 lanes
Audio	Limited	Yes
RS232/422/485	Yes	Yes
on board JTAG debugging	Yes	No
on board console to USB	Yes	Yes
on board bus sniffing	Limited	Limited

Peripherals

Memory

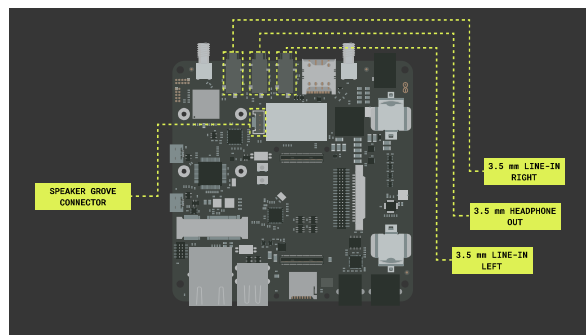
The Portenta Max Carrier equips two different memory units onboard, a flash memory and a mini SD card slot. The Flash memory onboard the Portenta Max Carrier has 2 MB of storage via QSPI. The Mini SD card interface makes it possible to extend the storage size. It can be used to process log data, from sensors or programmed on-board computer registry.

If you have an SD card connected to the Max Carrier you can create a directory on the SD card by using the following command:

```
1 mkdir -p /tmp/sdcard
```

Audio Interfaces

The Portenta Max Carrier features the CS42L52 from Cirrus Logic®, a stereo CODEC. The CS42L52 is a 24-bit, low-power stereo CODEC that can provide up to 1W per channel of Class D stereo/mono amplification to external speakers or enough power to drive 44mW per channel into stereo headphones. There are four analog audio interfaces on the Portenta Max Carrier, these are marked in the image below.



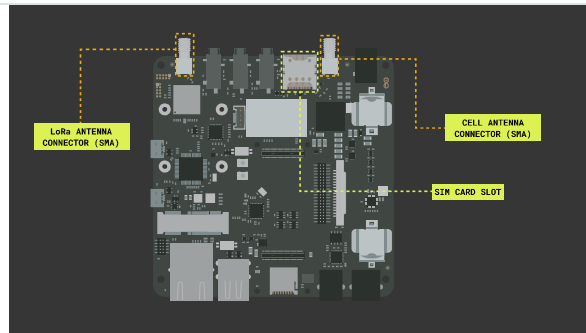
To use this feature with Linux, you could use something like the [alsa-lib](#). You can run it with this command:

```
1 apk update && apk add alsa-utils alsa-util
```

Connectivity

GSM Modem

The Portenta Max Carrier carries a cellular modem SARA-R412M-02B to carry out tasks requiring general network connectivity. This cellular modem is capable of establishing 2G / Cat-M1 / NB-IoT connections globally. It is powered by a bidirectional logic level shifter SN74LVC1T45 and an internal regulator, implemented for the use of SIM cards and I/O. The cellular modem requires a SIM card and an antenna connected to the SMA connector.



If you want to use this feature with Python® scripts, have a look at the [Modem Manager API](#).

LoRa® Technology

One of the many features of the Portenta Max Carrier is the Murata CMWX1ZZABZ-078 which enables LoRa® connectivity. LoRa technology is a Low Power Wide Area Network (LPWAN) protocol designed to connect low-power devices to the Internet. It was developed to meet and fulfill Internet of Things (IoT) devices' requirements, such as low-power consumption and low data throughput.

The appropriate antenna and frequencies depend on the region. The common frequencies are 915 MHz for North America and Australia and 863 MHz for the European region. Frequencies are on a range, so for example in the Australia region it is possible to use a 928 MHz compatible antenna and configuration.

For more in-depth information about LoRa® and its networks, please refer to [this documentation](#).

Ethernet

The Gigabit Ethernet physical interface is directly connected to the high-density connector to the Portenta X8 board. The connector includes an LED for indicating activity using the color orange, there is also a LED using the color green to indicate speed.

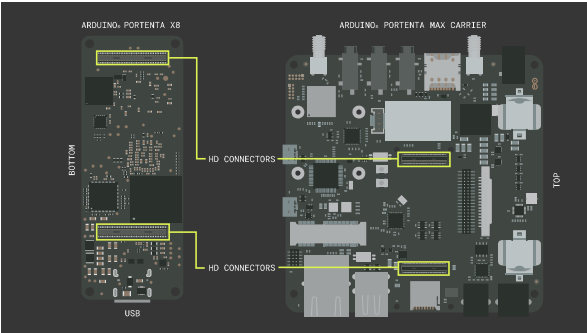
To access the 1 Gbps connection Ethernet peripheral on the Max Carrier you can use the following command:

```
1 ETH_1G=`dmesg | grep "fec 30be0000.ethernet
```

Instructions

Circuit

The Portenta Max Carrier and the Portenta X8 are connected with High Density connectors. The illustration below shows how to correctly connect the Portenta X8 and Portenta Max Carrier.



Setting up Portenta X8

If you wish to use the Portenta X8 and Max Carrier with the Arduino IDE, please follow our [Uploading sketches to Arduino Portenta X8 tutorial](#). Also, make sure that you have the latest version of the Portenta Core installed. Go to **Tools > Boards > board manager...** and search for **Arduino Mbed OS Portenta boards**. When this is installed and your Portenta X8 board is connected to the computer, the board should be assigned a port in the Arduino IDE.

If you want to use the Portenta X8 and Max Carrier with a FoundriesFactory, Python® scripts and Linux containers, Please take a look at our [Portenta X8 User Manual](#). You may also find our tutorial on how to [manage dockers on the Portenta X8](#) useful.

Quick Reference Tables for the Portenta Max Carrier

Peripheral Table

PERIPHERAL	PIN	FUNCTION	TYPE	DESCRIPTION
LoRa® Header				
	1	+3V3	Power	+3V3 Power Rail
	2	LoRa_SWDIO	Digital	LoRa® SWD Data Line
	3, 5, 9	GND	Power	Ground

PERIPHERAL	PIN	FUNCTION	TYPE	DESCRIPTION
	6 ~ 8	NC	NC	Not Connected
	4	LoRa_SWCLK	Digital	LoRa® SWD Clock Line
	10	LORA_RST	Digital	LoRa® module reset pin
Debug Header				
	1	3V3_DBG	Power	+3V3 Power Rail
	2	DBG_SWDIO	Digital	SWD Data Line
	3, 5, 9	GND	Power	Ground
	4	DBG_SWCLK	Digital	SWD Clock Line
	6 ~ 8	NC	NC	Not Connected
	10	RST	Digital	Reset Pin
JTAG Header				
	1	3V3	Power	+3V3 Power Rail
	2	JTAG_SWD	Digital	JTAG SWD Data Line
	3, 5	GND	Power	Ground
	4	JTAG_SCK	Digital	JTAG Clock Line
	6	JTAG_SWO	Digital	JTAG Trace Data Line
	7	NC	NC	Not Connected
	8	JTAG_TDI	Digital	JTAG Test Data In Line
	9	JTAG_TRST	Digital	JTAG Test Reset Line
	10	JTAG_RST	Digital	JTAG Reset Line
Header Connector				
	1	3V3	Power	+3V3 Power Rail
	2, 13, 14	GND	Power	Ground
	3	VREF_P	Analog	Positive Analog Voltage Reference
	4	VREF_N	Analog	Negative Analog Voltage

PERIPHERAL	PIN	FUNCTION	TYPE	DESCRIPTION
	5 ~ 12	A0 ~ A7	Analog	Analog input 0 ~ 7
	15	SAI_CLK	Digital	SAI Clock Line
	16	CS	Digital	SPI1 Clock Select
	17	SAI_FS	Digital	SAI Frame Synchronization
	18	CK	Digital	SPI1 Clock Line
	19	SAI_D0	Digital	SAI Data Line 0
	20	MISO	Digital	SPI Main In Secondary Out
	21	SAI_D1	Digital	SAI Data Line 1
	22	MOSI	Digital	SPI1 Main Out Secondary In
	23	SPDIF_TX	Digital	SPDIF Transmission
	24	TX	Digital	Serial3 Transmission
	25	SPDIF_RX	Digital	SPDIF Receive
	26	RX	Digital	Serial3 Receive
	27	SDA	Digital	I2C1 Data Line
	28	RTS	Digital	Serial3 Ready to Send
	29	SCL	Digital	I2C1 Clock Line
	30	CTS	Digital	Serial3 Clear to Send
	31	3V8	Power	+3V8 Power Rail
	32	SDA	Digital	I2C2 Data Line
	33	PWM0	Digital	PWM Output 0
	34	SCL	Digital	I2C2 Clock Line
	35	PWM1	Digital	PWM Output 1
	36	5V	Power	+5V Power Rail
	37	PWM2	Digital	PWM Output 2
	38	CAN1_TX	Digital	CAN Transmission Line
	39	PWM3	Digital	PWM Output 3
	40	CAN1_RX	Digital	CAN Receive Line
4P4C CAN Connector				

PERIPHERAL	PIN	FUNCTION	TYPE	DESCRIPTION
	2	CANL	Diff.	CAN Differential Signal Low
	3	CANH	Diff.	CAN Differential Signal High
	4	GND	Power	Ground
6P6C Serial Conn.				
	1	GND	Power	Ground
	2	Z	Digital	SP335 Out 1
	3	Y	Digital	SP335 Out 2
	4	A	Digital	SP335 In 2
	5	B	Digital	SP335 In 1
	6	GND	Power	Ground
PCIe Connector				
	1, 3, 5, 6, 8, 10, 12, 14, 16, 17, 19, 20, 24, 28, 37, 42 ~ 49, 51	NC	NC	Not Connected
	2, 39, 41, 52	3V3_PCIE	Power	+3V3 Power Rail
	3, 9, 15, 18, 21, 26	GND	Power	Ground

PERIPHERAL	PIN	FUNCTION	TYPE	DESCRIPTION
	29, 34, 35, 40, 50			
	7	PCIE_CLKREQ	Digital	PCIE Clock Request Signal
	11	mPCle_CK_N	Diff.	PCle Clock Differential Negative
	13	mPCle_CK_P	Diff.	PCle Clock Differential Positive
	22	mPCle_RST	Digital	PCle Reset Signal
	23	mPCle_RX_N	Diff.	PCle Receive Differential Negative
	25	mPCle_RX_P	Diff.	PCle Receive Differential Positive
	30	SCL	Digital	I2C Clock Line
	31	mPCle_TX_N	Diff.	PCle Transmission Differential Negative
	32	SDA	Digital	I2C Data Line
	33	mPCle_TX_P	Diff.	PCle Transmission Differential Positive
	36	USB_DN	Diff.	USB Data Differential Negative
	38	USB_DP	Diff.	USB Data Differential Positive

Trademark Acknowledgments

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